

PORTFOLIO

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Power Platform Developer

Case studies: Power Apps, Dataverse, Power Automate, Power BI, SharePoint, Azure AD (MSAL) & Microsoft Graph projects

This document summarizes eight internal enterprise applications and reports I designed and built end-to-end for a financial services organization — five Power Platform apps and three Power BI reporting dashboards. Each write-up covers the business problem, my role, the technical architecture, and the measured impact — without exposing the employer's name, live URLs, real employee data, or any information that could be commercially sensitive.

Live application links, real screenshots with employee data, and source code are intentionally withheld or redacted here; they can be shared selectively (e.g. a live walkthrough) during an interview upon request.

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APPENDIX

- A Note on Confidentiality

1. Digital Signature System

Paperless document signing & branch authorization platform

Problem

Branch staff at a multi-state credit co-operative society (100+ branches) relied on printing, physically signing, and couriering authorization documents to Head Office — slow, hard to track, and with no central visibility into which branches or devices were authorized to transact. IT/Establishment had no single system to manage branch authorizations or verify that a signed document actually came from an approved device.

My Role

Sole developer. Designed and built both halves of the system from scratch: the Branch Portal (used by branch staff) and the Admin Panel (used by IT/Establishment).

Architecture

Built as two independent client-side web apps with no backend server — SharePoint Online (via Microsoft Graph API) serves as the entire data layer, with Azure AD (MSAL) handling authentication for both portals.

Layer	Technology
Frontend	HTML, CSS, JavaScript (no framework, no build step)
Authentication	Microsoft MSAL (Azure AD)
Database	SharePoint Lists & Libraries via Microsoft Graph API
PDF Handling	Client-side PDF render, sign, and download
Hosting	Static site (GitHub Pages / any HTTPS host)

Note — why static hosting:

This app is deployed as a static site (GitHub Pages / any HTTPS host) rather than to a managed Azure service, because the organization did not provision paid Azure hosting (e.g. an Azure Function App or Azure Static Web Apps). With no backend budget available, the entire app was architected as client-side only, using SharePoint + Microsoft Graph as the data layer so a free static host is sufficient.

Key Features

- Login via Microsoft 365 SSO or Employee ID + Password, with first-time/permanent password setup
- Device / IP authorization check before granting branch-level access
- View, preview, and digitally sign pending documents (draw or upload signature)
- Personal signing history log for every user
- Idle-timeout auto logout for security
- Admin: branch management (add/edit/close, transfer branch managers), bulk CSV branch upload with progress tracking
- Admin: full branch "document journey" timeline, exportable to CSV/PDF
- Admin: document verification queue and branch-wise reporting (Excel/PDF export)

Impact

- Eliminated a fully paper-based signature/authorization process across 100+ branches
- Gave IT/Establishment centralized, auditable control over which branches and devices can sign documents
- Every signature now has a timestamped, traceable history — previously unavailable with the paper process

2. Secure Internal Communication Platform

Auditable branch-to-branch messaging, replacing WhatsApp

Problem

Branch-to-branch coordination was happening over WhatsApp — a consumer app with no audit trail, no access control tied to employment status, and organizational data (project updates, files, financial figures) leaving the company's security boundary entirely.

My Role

Sole developer, end-to-end: UI, authentication integration, and the SharePoint-backed data layer.

Architecture

Layer	Technology
Frontend	HTML, CSS, JavaScript
Authentication	Microsoft MSAL (Azure AD) — organizational account required
Database	SharePoint Lists via Microsoft Graph API
Hosting	GitHub Pages

Note — why static hosting:

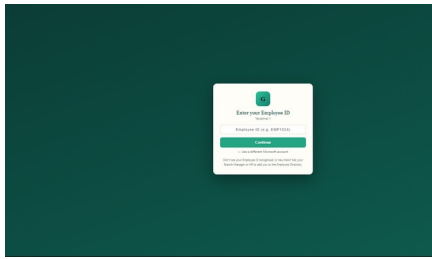
Like the Digital Signature System, this app is hosted as a static site (GitHub Pages / any HTTPS host) because the organization did not provide a paid Azure hosting service such as a Function App or Static Web Apps. The app is fully client-side, so a static host is all that's needed.

Key Features

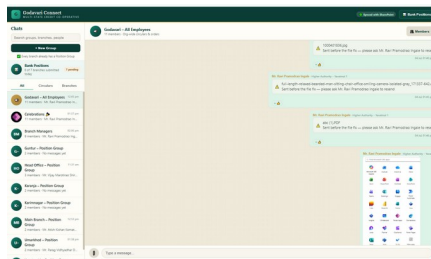
- Branch-to-branch messaging and file sharing inside a controlled, single-sign-on environment
- Dashboard with chat overview and unread/activity summary
- Approval workflow module for routing requests through the platform
- Position/status tracking module for operational figures relevant to daily branch reporting
- User profile management

Impact

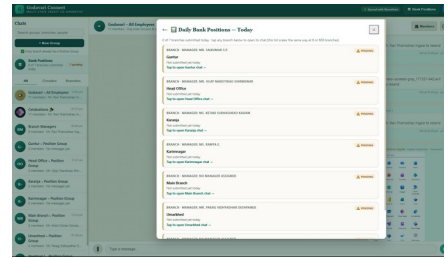
- Moved sensitive internal coordination out of a consumer chat app and inside the Microsoft 365 security boundary
- Introduced an auditable record of branch communications that didn't exist before
- Access is automatically revoked when an employee's organizational account is deactivated — unlike WhatsApp



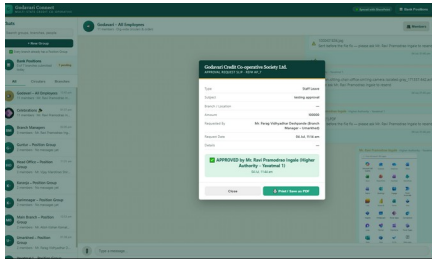
Sign-in — Employee ID (demo data)



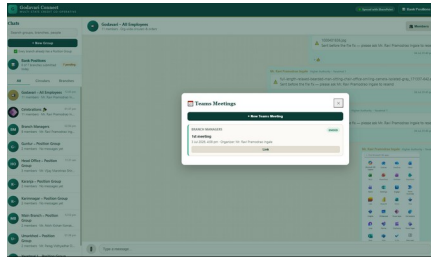
Chats dashboard overview (demo data)



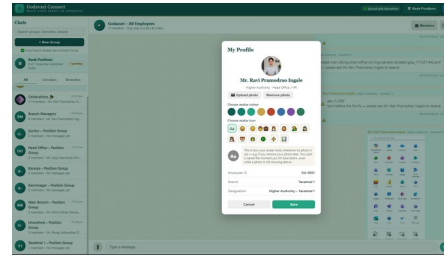
Daily Bank Positions module (demo data)



Approval request slip (demo data)



Teams Meetings module (demo data)



My Profile screen (demo data)

3. Organization Project Manager Portal

Central tracker for cross-team Power Platform initiatives

Problem

With multiple Power Apps builds, dashboards, and cross-team initiatives running in parallel, there was no single place for leadership to see project status, deadlines, or ownership — status updates were being gathered manually team by team.

My Role

Sole developer, end-to-end: information architecture, role-based views, and the reporting/export layer.

Architecture

Layer	Technology
Frontend	HTML, CSS, JavaScript
Authentication	Microsoft MSAL (Azure AD)
Database	SharePoint Lists via Microsoft Graph API
Hosting	GitHub Pages

Note — why static hosting:

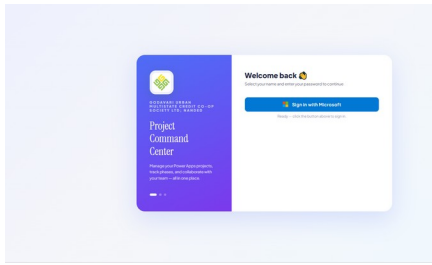
As with the two apps above, this portal is hosted as a static site (GitHub Pages / any HTTPS host) since the organization did not provide a paid Azure hosting service (Function App / Static Web Apps). SharePoint + Graph handles all data, so no server-side hosting is required.

Key Features

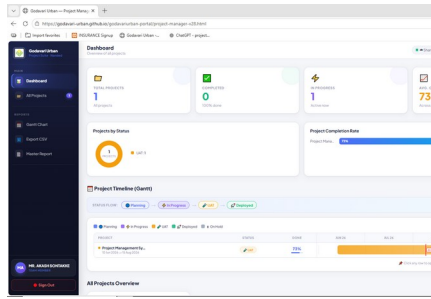
- Role-based access: Admin, Team Member, and Higher Authority (view-only)
- Live project tracking with progress %, deadlines, and overdue alerts
- Gantt-style timeline dashboard for at-a-glance scheduling
- Master report view with CSV export for offline analysis
- Dark mode support

Impact

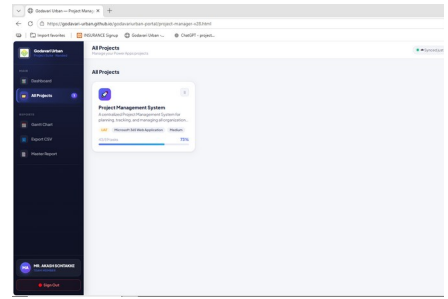
- Gave leadership self-serve visibility into project status instead of manual reporting cycles
- Overdue-alert logic surfaced at-risk projects earlier than the previous manual process
- Centralized reporting replaced scattered spreadsheets across teams



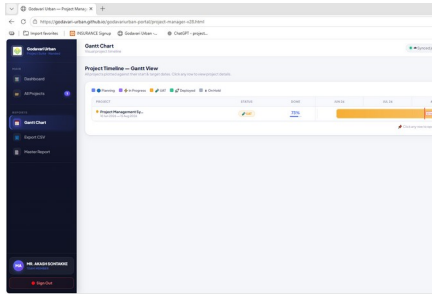
Sign-in screen (demo data)



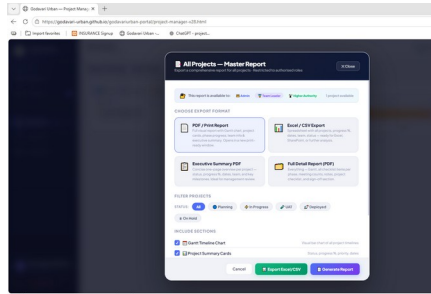
Dashboard overview (demo data)



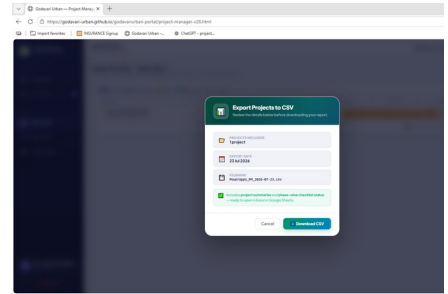
All Projects view (demo data)



Gantt chart timeline (demo data)



All Projects — Master Report export (demo data)



Export Projects to CSV (demo data)

4. FD Commission & Payout Management App

Branch-level fixed-deposit commission tracking with approval workflow

Problem

Fixed-deposit (FD) commission and payout requests submitted by branches were tracked manually, with no consistent calculation method, no approval trail for finance, and no daily consolidated view of submissions across branches.

My Role

Sole developer. Built the Canvas App, the commission-calculation logic, and the Power Automate approval flow.

Architecture

Layer	Technology
Frontend	Power Apps (Canvas App)
Backend / Data	SharePoint Online
Automation	Power Automate approval workflow
Logic	Power Fx

Key Features

- Branch-based login with role-based visibility of requests
- FD payout request entry with built-in commission calculation
- Daily submission summary view for supervisors
- Automated Power Automate approval routing

Impact

- Standardized commission calculation across all branches, removing manual/inconsistent computation
- Introduced a traceable, auditable approval trail for every payout request
- Gave supervisors a daily consolidated view instead of collecting figures branch by branch



Login page (demo data)

AGENT NAME	BL CODE	AC NO	PERIOD	COMMISSION %	COMMISSION AMT	CR ENTRY STATUS	STATUS	VERIFIED DATE	VERIFIED
Akash Sontakke	14	3104080	3000	4%	4000	DONE	Approved	12/16/2023	Yes
Yash Sontakke	14	3104045	100	4%	4000	NOT TAKEN	Rejected	12/16/2023	No
Yash Sontakke	14	3104080	3000	4%	4000	DONE	Approved	12/16/2023	Yes
Yash Sontakke	14	3104080	3000	4%	1400	NOT TAKEN	Rejected	12/16/2023	No
Yash Sontakke	14	3104080	3000	4%	1400	DONE	Approved	12/16/2023	Yes
Akash Sontakke	14	3104080	3000	4%	14000		Approved	11/23/2023	Yes
Prakash Sontakke	14	3104048	3000	5%	23000		Approved	11/23/2023	Yes
Yash Sontakke	14	3104080	3000	4%	8000		Approved	11/23/2023	Yes

Head Office — HO Payout Entries / Approval (demo data)

AGENT NAME	CR ENTRY STATUS	BL CODE	AC NO	PERIOD	COMMISSION %	COMMISSION AMT	STATUS	VERIFIED DATE
Yash Sontakke	APPROVED	14	3104080	3000	4%	4000	Approved	12/16/2023
Akash Sontakke	APPROVED	14	3104080	3000	4%	4000	Approved	12/16/2023
Yash Sontakke	NOT TAKEN	14	3104045	100	4%	4000	Rejected	

Branches — Payout Entries (demo data)

5. Automated Leave Management System

End-to-end digital leave request & approval portal

Problem

Leave requests across 100+ branches were tracked manually with no automated approval routing, no real-time status visibility for employees, and frequent data-entry errors from re-typing employee details by hand.

My Role

Power Platform Developer. Designed the data model, built the Canvas App UI, and authored the Power Automate approval engine.

Architecture

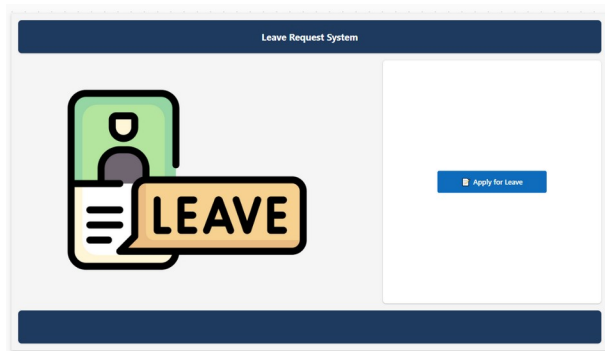
Layer	Technology
Frontend	Power Apps (Canvas App)
Backend / Data	Microsoft Dataverse — relational tables with filtering logic for real-time, user-specific views
Automation	Power Automate cloud flow for multi-stage Approve/Reject routing and email notifications
Identity	Microsoft 365 (User()) function for auto-login and profile photo integration

Key Features

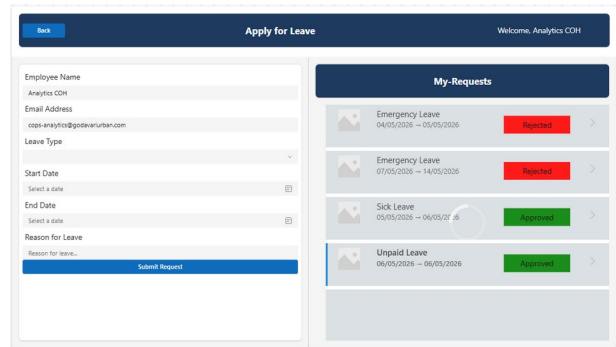
- Automatic login detection and personalized access using the signed-in Microsoft 365 identity
- Multi-stage manager approval workflow with automated email notifications on Approve/Reject
- Real-time "Ticket Journey" dashboard so employees can track their own request status without contacting HR
- Auto-populated employee details pulled directly from the organization's directory, reducing manual entry
- High-contrast, modern UI (glassmorphism/neumorphism styling) designed for non-technical branch staff
- Collections + Refresh()-based data loading tuned for performance under concurrent branch usage

Impact

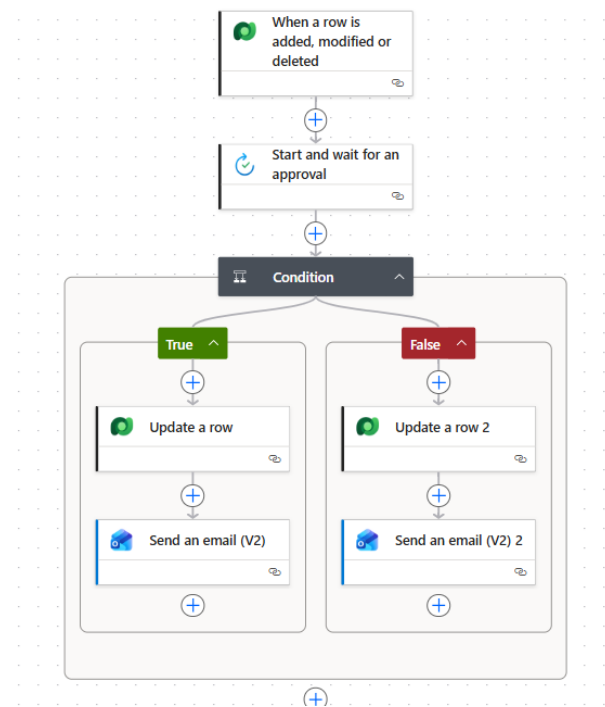
- Removed paper-based leave requests across the entire multi-branch organization
- Gave employees self-serve visibility into approval status, reducing HR's manual status-update workload
- Reduced data-entry errors by auto-populating employee details instead of manual input



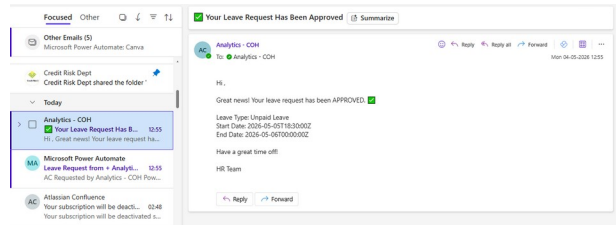
Leave Request System — home screen (demo data)



Apply for Leave & My-Requests screen (demo data)



Power Automate approval workflow (demo data)



Automated approval email (demo data)

6. Credit Control & Recovery Dashboard

Power BI reporting suite for loan recovery, collections & field-visit performance

Problem

The Credit Control department at a multi-state credit co-operative society (100+ branches) needed a consolidated way to track loan recovery performance across all branches and loan types (vehicle, housing, personal, cash credit, micro-finance, etc.), alongside the productivity of the telecalling and field-visit (CFO) recovery teams. This data previously lived across separate exports with no unified, drillable view for management.

My Role

Sole developer. Modeled the data, built all DAX measures, and designed the full multi-page report suite end-to-end.

Architecture

Layer	Technology
Tool	Power BI Desktop
Data Modeling	Power Query (M) for transformation, star-schema data model
Calculations	DAX measures for recovery totals, OTS (One-Time Settlement) tracking, call/visit productivity
Sources	Internal loan/collections data exports, consolidated and refreshed into the model

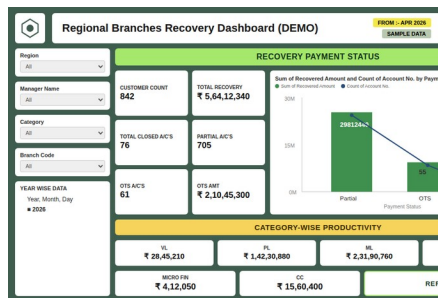
Key Features

- Executive dashboard: total recovery, customer counts, OTS/regular/partial/closed account status, latest call & visit dates
- Branch and loan-type productivity views broken down by product (vehicle, housing, personal, cash credit, warehouse, micro-finance loans)
- Telecalling analytics: answered / unanswered / disconnected call volumes, daily and monthly trends
- Field-visit (CFO) tracking: visit counts, task completion status, visits created by telecallers vs. field staff
- Daily and year-wise branch recovery reports with account-closure and recovery-amount trends
- Loan GL balance summary by account type and branch
- Interactive slicers for year, branch, and loan-type filtering across every page

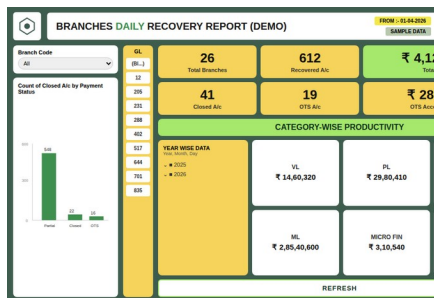
Impact

- Gave management a single, drillable view of recovery performance instead of manually consolidating branch-level exports
- Made telecalling and field-visit productivity visible and comparable across branches for the first time
- Daily recovery reporting replaced a previously manual, spreadsheet-based process

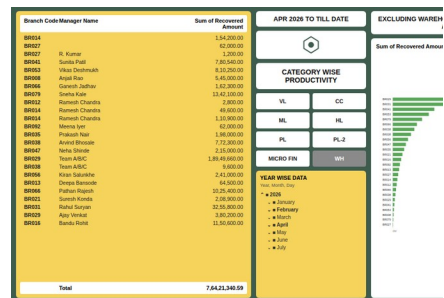
Report Preview



Daily recovery overview (demo data)



Regional recovery dashboard (demo data)



Branch/manager recovery detail (demo data)

7. Insurance Policy Management Dashboard

Power BI reporting for customer loan insurance, vehicle & contingency policies

Problem

Insurance policies tied to customer loans, company vehicles, employee/director coverage, and contingency policies were tracked with no central reporting layer — making it hard to see policy renewal dates, coverage counts by branch, or exposure by insurance company and policy type at a glance.

My Role

Sole developer. Built the data model and full multi-page report from raw policy records.

Architecture

Layer	Technology
Tool	Power BI Desktop
Data Modeling	Power Query (M) for transformation, DAX for aggregated policy metrics
Sources	Internal policy/insurance records, consolidated and refreshed into the model

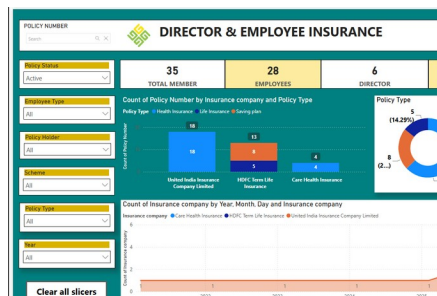
Key Features

- Customer loan insurance: branch-wise policy counts, GL-wise policy breakdown, top-10 branch ranking, full policy detail table
- Director & employee coverage dashboard: policy counts by insurance company and policy type
- Vehicle insurance tracking with renewal trend visuals
- Contingency policy tracking: firm policy renewal dates, sum-insured breakdown by policy sub-type
- Company-wide policy-type distribution and searchable policy-number lookup

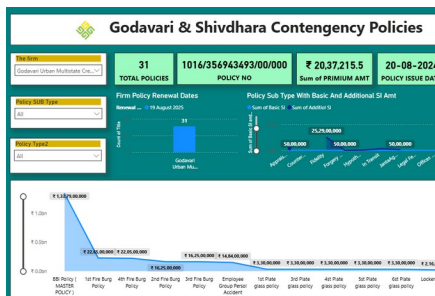
Impact

- Replaced scattered policy records with a single reportable view across all policy categories
- Made upcoming renewal dates visible in advance, reducing the risk of lapsed coverage
- Gave leadership a branch-level and company-level view of insurance exposure that didn't exist before

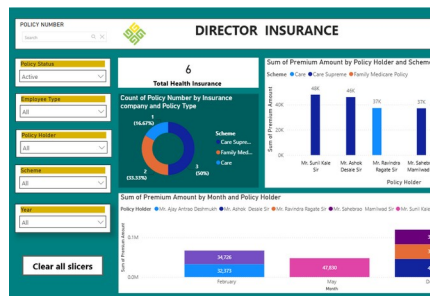
Report Preview



Director & Employee Insurance overview (demo data)



Contingency policies tracking (demo data)



Director Insurance dashboard (demo data)

8. Head Office Daily Productivity Report

Power BI dashboard for department-wise, employee-level daily output tracking

Problem

Head Office departments (Credit, HMPL, Accounts Payable & Insurance, and the Recovery/Reconciliation desk) had no consistent way to see individual employee output on a given day — work was logged informally, with no manager-level visibility into volumes handled per employee or per department.

My Role

Sole developer. Built the data model and single-page productivity report from daily work-log entries.

Architecture

Layer	Technology
Tool	Power BI Desktop
Data Modeling	Power Query (M) for transformation, DAX for daily aggregation
Sources	Daily work-log entries consolidated per department and refreshed into the model

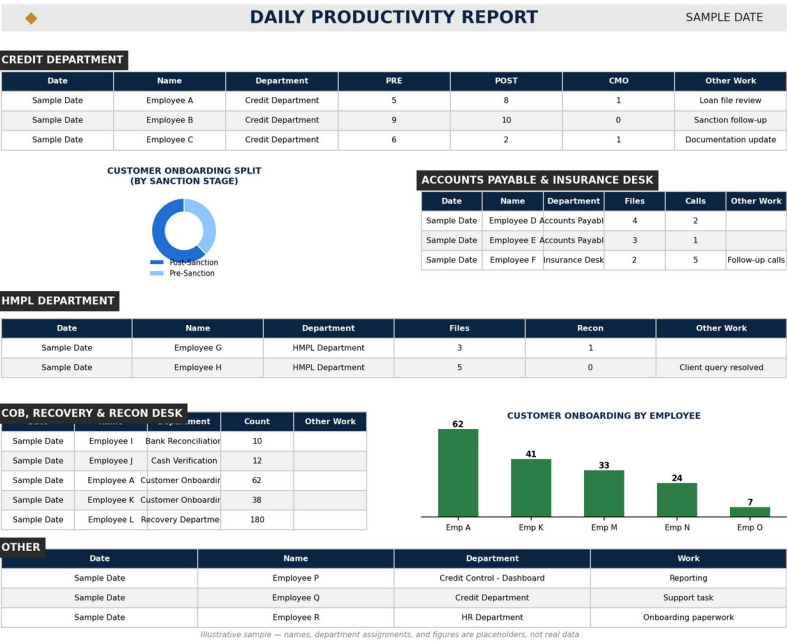
Key Features

- Department-wise employee productivity tables for Credit, HMPL, Accounts Payable & Insurance, and Recovery/Reconciliation desks
- Employee-wise breakdown of pre-processing vs. post-processing item counts, calls handled, and files processed
- Customer-onboarding count by employee, with a column chart ranking daily output
- Pre- vs. post-processing split shown as a donut chart for a quick daily balance check
- Date-based filtering so any single day's productivity can be reviewed department by department

Impact

- Gave department heads a same-day view of individual and team output instead of end-of-week manual summaries
- Made uneven workload distribution across employees visible for the first time
- Replaced informal, register-based daily reporting with a consistent, comparable digital record

Report Preview



Layout matches the live report; names, department assignments, and figures shown are illustrative placeholders, not real employee data.

A Note on Confidentiality

These case studies describe the systems I designed and built, generalized to protect my employer's identity and any operationally sensitive details (organization name, live URLs, internal data, branch counts beyond a general range). No real employee data, credentials, tenant IDs, or internal SharePoint site information appears in this document or should appear in any screenshots inserted into it.

If sharing this alongside GitHub: repositories still named after or linked to the organization (via URL slugs, demo video links, or license text) should be renamed or made private, since a case study here doesn't remove information already public elsewhere.